

Frequent Pattern

A pattern that appears frequently in a dataset

Types of Frequent Patterns:

Itemset

Set of items bought together
Example: {milk, bread}

Sequential Pattern:

Items bought in a sequence over time
Example: buy phone → then buy earphones

Market Basket Analysis

Technique to analyze customer purchase behavior AND

Finds relationships between products

Example:

Transactions:

- T1: {milk, bread, butter}
- T2: {milk, biscuits}
- T3: {bread, butter}

Rule:

Milk → Bread

Benefits :

- **Improves product placement**
- **Increases sales (cross-selling)**
- **Helps recommendation systems**
- **Better customer understanding**
- **Supports marketing strategies**

Techniques / Methods

Apriori Algorithm

- Finds frequent itemsets step-by-step
- Uses support threshold

FP-Growth Algorithm

- Faster than Apriori
- Uses tree structure

Association Rules

- Uses **Support & Confidence**

Real-Life Scenario

Step 1: Count Single Items

- milk $\rightarrow 8/10 = 80\%$
- bread $\rightarrow 8/10 = 80\%$
- eggs $\rightarrow 5/10 = 50\%$

Step 2: Pair Itemsets

- {milk, bread} $\rightarrow 6/10 = 60\%$
- {milk, eggs} $\rightarrow 4/10 = 40\%$
- {bread, eggs} $\rightarrow 4/10 = 40\%$

Final Frequent Itemsets (>50%)

- milk
- bread
- eggs
- {milk, bread}

Only **ONE** strong pair above 50%

Transactions (10 total):

- T1: {milk, bread, eggs}
- T2: {milk, bread}
- T3: {milk, bread, butter}
- T4: {milk, eggs}
- T5: {bread, eggs}
- T6: {milk, bread, eggs}
- T7: {milk, bread}
- T8: {bread, butter}
- T9: {milk, bread, eggs}
- T10: {milk, juice}

Minimum Support = **50%**